

Why we Use Code First rather than Database First?

Code First is often preferred because it allows developers to design and control the application's data model directly from the code. Instead of starting with an existing database, the developer creates classes that represent the domain model, and the database is automatically generated from these classes using Entity Framework migrations.

Code First improves productivity because developers can modify the model easily in the code and update the database through migrations without manually editing database schemas. It also integrates naturally with object-oriented programming principles, making the code easier to maintain and evolve. Additionally, it supports version control for database changes through migrations, which helps teams track and manage schema updates during development.

In contrast, Database First is mainly useful when a database already exists and must be used as it is.

Why we use Fluent API rather than Data Annotations?

Fluent API provides more flexibility and control over how entities are mapped to the database compared to Data Annotations. While Data Annotations use attributes directly inside the model classes, Fluent API allows configuration in the DbContext using method chaining.

One advantage of Fluent API is that it keeps the domain classes clean and free from database-specific attributes. It also supports advanced configurations that cannot be achieved using Data Annotations alone, such as complex relationships, composite keys, detailed column configurations, and precise control over entity mappings.

Therefore, Fluent API is often preferred in larger or more complex applications where detailed configuration and separation between business logic and database mapping are important.